High Dimensionality

Summary sheet — Deep Learning (S.-A. Berrani, ENSIA 2026–2027)

Core idea & notation

High-dimensional data (dimension > 3) escapes our intuition: distances, volumes and sample requirements behave in counter-intuitive ways. All these difficulties are called the **dimensionality curse** (Bellman, 1961). Analysis, exploration, processing and management all become harder.

- d — dimension (number of variables); n — number of individuals (samples).
- $V_d(r)$ — volume of a d-dimensional hyper-sphere of radius r; ε — width of the frame in the hyper-cube.
- $D_{\min}, D_{\max}$ — distance from a query point to its nearest / furthest neighbour.

1. The dimensionality curse

Origin (Bellman): difficulty of optimizing in high-dimensional spaces with *exhaustive* methods. **Statistics (Donoho, 2000):** link between sample size and reliability of an estimate.

Property: to keep the same reliability, the number of samples must grow *exponentially* with d.

Identification. More variables *and* more individuals $\Rightarrow$ less intuition, costly processing, harder analysis. Whenever a method needs "enough points in every region of space", expect the curse.

1.1 Quantities that grow exponentially with d

Complementary examples (from external sources, not in the slides):

Quantity	Growth	Illustration
Grid points to cover $[0,1]^d$ with spacing δ	$(1/\delta)^d$	$\delta = 0.01$: 10^2 points for $d=1$, 10^6 for $d=3$, 10^{20} for $d=10$
Samples for error $\leq \varepsilon$ (Lipschitz L)	$\approx (L/2\varepsilon)^d$	$\varepsilon = 0.1$, $d=100$: $(5L)^{100}$ samples
Binary combinations of d variables	2^d	each new dimension doubles the effort of trying all cases
Cells to inspect around a point (grid NN search)	3^d	9 in 2D, 27 in 3D
Vertices of the hyper-cube	2^d	corners hold almost all the volume

2. Loss of the notion of distance

When d increases (data uniform and independent on every dimension, Beyer 1999):

- the distance to the **nearest** neighbour increases;
- the distance to the **furthest** neighbour increases;
- but $D_{\min}$ increases **faster** than $D_{\max}$.

Limit:

$$d \to \infty \implies \boxed{D_{\min} \approx D_{\max}} \qquad \left(\text{i.e. } \frac{D_{\max}}{D_{\min}} \to 1\right)$$

Consequence. Nearest and furthest neighbours become indistinguishable: all points look equally far apart, so distance-based methods (nearest neighbour, clustering with Euclidean distance) lose meaning.

3. Volume of the hyper-sphere

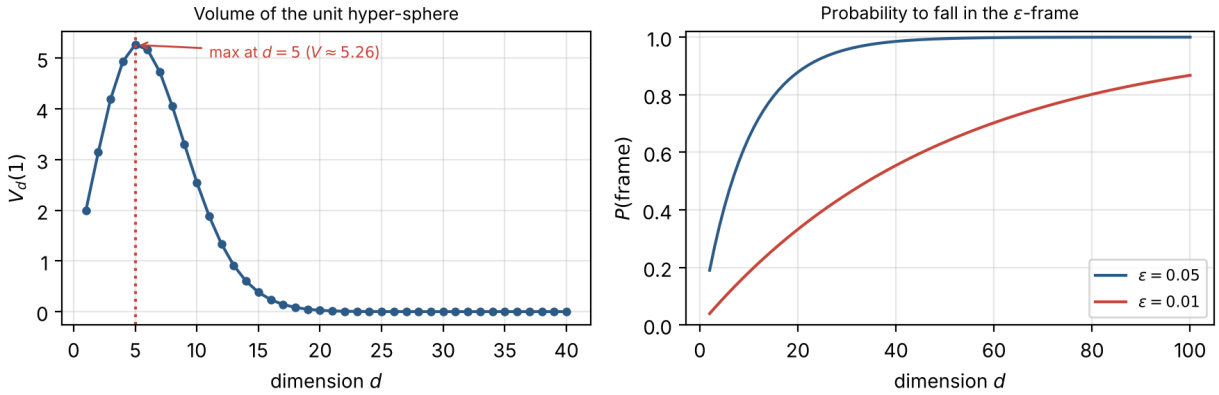
Formula:

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma\left(\frac{d}{2}+1\right)} r^d$$

Example: unit sphere, $r = 1$

d	2	3	5	10	20	50	100
$V_d(1)$	3.14	4.19	**5.26**	2.55	2.6×10^{-2}	1.7×10^{-13}	2.4×10^{-40}
$V_d(1)/2^d$ (sphere / enclosing cube)	0.785	0.524	0.164	2.5×10^{-3}	2.5×10^{-8}	10^{-28}	10^{-70}

The volume rises up to $d = 5$, then collapses to 0; relative to the cube, the sphere becomes negligible: *the volume of the cube concentrates in its corners.*



Left: volume of the unit sphere vs d (peak at $d = 5$). Right: probability to fall in the ε-frame of the hyper-cube (exercise below).

4. Exercise: hyper-cube and its frame

Setup: points uniform in the hyper-cube $[0,1]^d$. The *frame* (grey) is the set of points at distance $\leq \varepsilon$ from the edge, $0 < \varepsilon < \frac{1}{2}$. The remaining inner hyper-cube is $[\varepsilon, 1-\varepsilon]^d$, of side $1-2\varepsilon$.

Proof. For a uniform distribution, probability = volume ratio:

$$P(\text{inner}) = \frac{\text{Vol}([\varepsilon, 1-\varepsilon]^d)}{\text{Vol}([0,1]^d)} = (1-2\varepsilon)^d, \qquad \boxed{P(\text{frame}) = 1-(1-2\varepsilon)^d}$$

Since $0 < 1-2\varepsilon < 1$: $(1-2\varepsilon)^d \to 0$, hence $P(\text{frame}) \to 1$ as $d \to \infty$.

Example: $\varepsilon = 0.05$ **(so** $1-2\varepsilon = 0.9$**)**

$$d = 2: \ 0.19, \quad d = 3: \ 0.27, \quad d = 10: \ 0.65, \quad d = 50: \ 0.995, \quad d = 100: \ 0.99997$$

Number of dimensions for $P(\text{frame}) \geq p$: $d \geq \dfrac{\ln(1-p)}{\ln(1-2\varepsilon)}$; for $p = 0.99$: $d \geq 43.7$, i.e. $d \geq 44$.

What we notice. The curve is increasing and saturates at 1: in high dimension, **almost every point lies near the edge** and the "centre" is empty. Data are sparse and isolated.

5. Solutions

	1. Reduce the variables	**2. Reduce the instances**
Idea	create new variables from the original ones while losing as little information as possible	partition the dataset by grouping similar individuals
Acts on	columns (features), $p \to q < p$	rows (individuals)
Example	**PCA** (Principal Component Analysis)	**Clustering** (unsupervised classification)

6. Quick-reference checklist

Phenomenon	Formula / fact	Consequence
Sampling	samples $\sim$ exponential in d	data become sparse
Distances	$D_{\min}$ grows faster than $D_{\max}$; $D_{\max}/D_{\min} \to 1$	nearest neighbour loses meaning
Sphere volume	$V_d = \pi^{d/2} r^d / \Gamma(d/2+1) \to 0$	sphere negligible vs cube
Frame	$P = 1 - (1-2\varepsilon)^d \to 1$	points near the edge

Exam reflex: (1) state the curse (Bellman / Donoho) $\to$ (2) exponential growth of samples and volume $\to$ (3) $D_{\min} \approx D_{\max} \to$ (4) $V_d(r)$ formula and sphere/cube ratio $\to$ (5) frame: $P = 1 - (1-2\varepsilon)^d \to 1 \to$ (6) fix: reduce features (PCA) or instances (clustering).